Technical University of Denmark

Advanced machine learning (02460)

Practice Exam Set A

Exam duration: 2 hours. Aid: None.

IMPORTANT: Questions are self-contained; all necessary formulas are provided.
Only information written on pages 2–11 will be evaluated.
Write your student number on all pages at the bottom-left.

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Student number:

Part I Generative models

Exercise A — Variational autoencoder (VAE)

Consider a deep latent variable model with latent variable $z \in \mathbb{R}$ and observed variable $\mathbf{x} \in \mathbb{R}^2$. Let z follow a standard Gaussian prior

$$p(z) = \mathcal{N}(z \mid 0, 1),$$

and let the likelihood be

$$p(\mathbf{x} \mid z) = \mathcal{N}\big(\mathbf{x} \mid [z, -z]^\top, \mathbf{I}_2\big),$$

where $\mathbf{I}_2 \in \mathbb{R}^{2\times 2}$ is the identity matrix. Consider the variational posterior

$$q(z \mid \mathbf{x}) = \mathcal{N}\big(z \mid x_1 + x_2,\ 1\big).$$

Recall that the ELBO is

$$\mathrm{ELBO}(\mathbf{x}) = \mathbb{E}_{z\sim q(z|\mathbf{x})}\big[\ln p(\mathbf{x} \mid z)\big] - \mathrm{KL}\big[q(z \mid \mathbf{x}) \parallel p(z)\big], \tag{1}$$

that the bivariate Gaussian density with identity covariance is

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{I}_2) = \tfrac{1}{2\pi} \exp\big(-\tfrac{1}{2}\|\mathbf{x} - \boldsymbol{\mu}\|^2\big), \tag{2}$$

and that the KL between two univariate Gaussians is

$$\mathrm{KL}\big[\mathcal{N}(\mu_1, \sigma_1^2) \parallel \mathcal{N}(\mu_2, \sigma_2^2)\big] = \ln\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}. \tag{3}$$

Question A.1: Assuming a one-sample Monte Carlo estimate of the reconstruction term with sampled value $z' = 1$, what is $\mathrm{ELBO}(\mathbf{x}')$ evaluated at $\mathbf{x}' = [2, -1]^\top$?

- **A** ☐ $\mathrm{ELBO}(\mathbf{x}') = -1 - \ln(2\pi)$
- **B** ☐ $\mathrm{ELBO}(\mathbf{x}') = -\frac{1}{2} - \ln(2\pi)$
- **C** ☐ $\mathrm{ELBO}(\mathbf{x}') = -\frac{3}{2} - \ln(2\pi)$
- **D** ☐ $\mathrm{ELBO}(\mathbf{x}') = \frac{1}{2} - \ln(2\pi)$

Give a brief argument for your answer.

Exercise B — Normalizing flows

Consider a normalizing flow with random variables $\mathbf{u}, \mathbf{x} \in \mathbb{R}^2$, where the base distribution is $p_{\mathbf{u}}(\mathbf{u}) = \mathcal{N}(\mathbf{u} \mid \mathbf{0}, \mathbf{I}_2)$, and the density of $\mathbf{x}$ is

$$p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{u}}(\mathbf{u}) \, | \det J_T(\mathbf{u})|^{-1}, \qquad \mathbf{u} = T^{-1}(\mathbf{x}). \tag{4}$$

Let $T = T_2 \circ T_1$ with the following known values:

$$\begin{aligned} T_1([0,0]^\top) &= [1,0]^\top, & \det J_{T_1}([0,0]^\top) &= 2, \\ T_1([2,0]^\top) &= [3,1]^\top, & \det J_{T_1}([2,0]^\top) &= 4, \\ T_2([1,0]^\top) &= [1,1]^\top, & \det J_{T_2}([1,0]^\top) &= 3, \\ T_2([3,1]^\top) &= [4,2]^\top, & \det J_{T_2}([3,1]^\top) &= 5. \end{aligned}$$

Recall that $(T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1}$ and

$$\det J_{T_2 \circ T_1}(\mathbf{u}) = \det J_{T_2}(T_1(\mathbf{u})) \cdot \det J_{T_1}(\mathbf{u}). \tag{5}$$

Question B.1: What is $p_{\mathbf{x}}([1,1]^\top)$?

- **A** ☐ $p_{\mathbf{x}}([1,1]^\top) = \dfrac{1}{6\pi}$
- **B** ☐ $p_{\mathbf{x}}([1,1]^\top) = \dfrac{1}{12\pi}$
- **C** ☐ $p_{\mathbf{x}}([1,1]^\top) = \dfrac{e^{-2}}{6\pi}$
- **D** ☐ $p_{\mathbf{x}}([1,1]^\top) = \dfrac{e^{-2}}{12\pi}$

Give a brief argument for your answer.

Exercise C — Denoising Diffusion Probabilistic Models

Consider a DDPM with $T = 5$ latent variables, where $\mathbf{x}_0 \in \mathbb{R}^2$ is the data and $\mathbf{x}_1, \ldots, \mathbf{x}_5 \in \mathbb{R}^2$ are the latent variables. Let the variance schedule be

$$\beta_1 = \tfrac{1}{2}, \quad \beta_2 = \tfrac{1}{2}, \quad \beta_3 = \tfrac{1}{4}, \quad \beta_4 = \tfrac{1}{4}, \quad \beta_5 = \tfrac{1}{4}.$$

Recall that the forward process satisfies

$$q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t \mid \mathbf{x}_{t-1}), \qquad q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\big(\mathbf{x}_t \mid \sqrt{1-\beta_t}\, \mathbf{x}_{t-1},\, \beta_t \mathbf{I}\big), \tag{6}$$

and that marginalising over $\mathbf{x}_{1:t-1}$ gives

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\big(\mathbf{x}_t \mid \sqrt{\bar{\alpha}_t}\, \mathbf{x}_0,\, (1-\bar{\alpha}_t)\mathbf{I}\big), \qquad \bar{\alpha}_t = \prod_{\tau=1}^{t} (1-\beta_\tau). \tag{7}$$

Question C.1: What is $q\big(\mathbf{x}_2 \mid \mathbf{x}_0 = [4, -4]^\top\big)$?

A ☐ $\mathcal{N}\big(\mathbf{x}_2 \mid [2, -2]^\top,\, \frac{3}{4}\mathbf{I}\big)$
B ☐ $\mathcal{N}\big(\mathbf{x}_2 \mid [2, -2]^\top,\, \frac{1}{4}\mathbf{I}\big)$
C ☐ $\mathcal{N}\big(\mathbf{x}_2 \mid [1, -1]^\top,\, \frac{3}{4}\mathbf{I}\big)$
D ☐ $\mathcal{N}\big(\mathbf{x}_2 \mid [1, -1]^\top,\, \frac{1}{4}\mathbf{I}\big)$

Give a brief argument for your answer.

Part II Representation learning

Exercise D — Pull-back metrics

Consider the generative model $f : \mathbb{R}^d \to \mathbb{R}^D$ $(D \geq d)$

$$f(\mathbf{x}) = W_2 \,\mathrm{ReLU}(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2, \tag{8}$$

where $W_1 \in \mathbb{R}^{D\times d}$ and $W_2 \in \mathbb{R}^{D\times D}$ are full-rank matrices, $\mathbf{b}_1 \in \mathbb{R}^D$, $\mathbf{b}_2 \in \mathbb{R}^D$, and $\mathrm{ReLU}(x) = \max(0, x)$ is applied element-wise. Define

$$\Lambda_1 = \mathrm{diag}\big(\mathbf{1}[W_1\mathbf{x} + \mathbf{b}_1 > \mathbf{0}]\big), \quad \Lambda_0 = \mathrm{diag}\big(\mathbf{1}[W_1\mathbf{x} > \mathbf{0}]\big), \tag{9}$$

where $\mathbf{1}[\cdot]$ is the element-wise indicator function.

The pull-back metric is $G_\mathbf{x} = J_\mathbf{x}^\top J_\mathbf{x}$, where $J_\mathbf{x}$ is the Jacobian of f. Which one of the following is correct?

A ☐ $G_\mathbf{x} = W_1^\top \Lambda_1 W_2^\top W_2 \Lambda_1 W_1$
B ☐ $G_\mathbf{x} = W_2^\top \Lambda_1 W_1^\top W_1 \Lambda_1 W_2$
C ☐ $G_\mathbf{x} = W_1^\top \Lambda_0 W_2^\top W_2 \Lambda_0 W_1$
D ☐ $G_\mathbf{x} = W_2^\top \Lambda_0 W_1^\top W_1 \Lambda_0 W_2$

Give a brief argument for your answer.

Exercise E — Curve energy

Consider the generative model $\mathbf{y} = f(\mathbf{x})$, where

$$f(\mathbf{x}) = \left[x_1^2,\ x_1 x_2,\ x_2\right]^\top. \tag{10}$$

Furthermore, consider the two latent curves

$$\mathbf{c}_1(t) = t[1,0]^\top \qquad \text{and} \qquad \mathbf{c}_2(t) = t[0,1]^\top, \quad t \in [0,1].$$

Question E.1: Derive the pull-back metric $G_\mathbf{x} = J_\mathbf{x}^\top J_\mathbf{x}$, where $J_\mathbf{x}$ is the Jacobian of f at $\mathbf{x}$.

Question E.2: The energy of a curve is defined as

$$E(\mathbf{c}) = \int_0^1 \dot{\mathbf{c}}(t)^\top G_{\mathbf{c}(t)}\, \dot{\mathbf{c}}(t)\, \mathrm{d}t, \tag{11}$$

where $\dot{\mathbf{c}}(t) = \mathrm{d}\mathbf{c}(t)/\mathrm{d}t$. Evaluate $E(\mathbf{c}_1)$ and $E(\mathbf{c}_2)$.

$E(\mathbf{c}_1) =$ $\qquad\qquad$ $E(\mathbf{c}_2) =$

Sketch your derivations below.

Exercise F — Riemannian integration

Consider the Riemannian metric over $\mathbf{x} = (x_1, x_2) \in [0,1] \times [0,1]$

$$G_{\mathbf{x}} = \mathrm{diag}(x_2^2,\ x_1^2). \tag{12}$$

Question F.1: Derive the volume measure $\mathrm{d}M(\mathbf{x}) = \sqrt{\det G_{\mathbf{x}}}\,\mathrm{d}\mathbf{x}$.

$$\mathrm{d}M(\mathbf{x}) =$$

Sketch your derivations below.

Question F.2: Let $\Omega = [0,1] \times [0,1]$. Evaluate the integrals

$$I_1 = \int_\Omega x_1^2\,\mathrm{d}M(\mathbf{x}), \qquad I_2 = \int_\Omega x_2\,\mathrm{d}M(\mathbf{x}).$$

$I_1 =$ $\qquad\qquad$ $I_2 =$

Sketch your derivations below.

Exercise G — ELBO and evidence

Recall that for a latent variable model with prior $p(\mathbf{z})$, likelihood $p(\mathbf{x} \mid \mathbf{z})$, and approximate posterior $q(\mathbf{z} \mid \mathbf{x})$, the ELBO is defined as

$$\mathrm{ELBO}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{z}|\mathbf{x})}\Big[\ln \frac{p(\mathbf{x}, \mathbf{z})}{q(\mathbf{z} \mid \mathbf{x})}\Big]. \tag{13}$$

Question G.1: Show that the ELBO can be written as

$$\mathrm{ELBO}(\mathbf{x}) = \ln p(\mathbf{x}) - \mathrm{KL}\big[q(\mathbf{z} \mid \mathbf{x}) \parallel p(\mathbf{z} \mid \mathbf{x})\big]. \tag{14}$$

Hint: Use the factorisation $p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z} \mid \mathbf{x})\, p(\mathbf{x})$.

Part III Graph models

Exercise H — Graph Laplacians

The random-walk graph Laplacian is defined as $L_{\text{rw}} = I - D^{-1}A$, where A is the adjacency matrix and D the degree matrix. Consider a graph with

$$L_{\text{rw}} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ -\frac{1}{2} & 1 & -\frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix},$$

with nodes labelled a–e in row order.

Question H.1: Sketch the graph.

Question H.2: Consider a random walk on this graph starting at node a. What is the probability p of being at node b after exactly 3 steps?

$$p =$$

Give a brief argument for your answer.

Exercise I — Weisfeiler-Lehman test

Consider the following graph with nodes a, b, c, d: a triangle formed by a–b–c–a, with an additional edge a–d (so d is a pendant connected only to a). Run the Weisfeiler-Lehman (WL) algorithm:

1. Assign initial labels $\ell_v^{(0)} = d_v$ (degree of node v).
2. Iteratively: $\ell_v^{(i)} = \text{hash}\big(\{\{\ell_u^{(i-1)} : u \in \mathcal{N}(v)\}\}\big)$.

Question I.1: Which of the following are true? Select all that apply.

A ☐ $\ell_a^{(0)} = 3$

B ☐ $\ell_d^{(0)} = 2$
C ☐ $\ell_b^{(1)} = \ell_c^{(1)}$
D ☐ $\ell_a^{(1)} = \ell_b^{(1)}$
E ☐ $\ell_d^{(1)} = \ell_b^{(1)}$

Question I.2: After how many rounds does the WL algorithm halt for this graph?

Number of rounds =

Give a brief argument.

Exercise J — Shallow embeddings

We use the squared-distance decoder $\mathrm{DEC}(\mathbf{z}_u, \mathbf{z}_v) = \|\mathbf{z}_u - \mathbf{z}_v\|^2$ in a 2-dimensional embedding space with $\mathbf{z}_v = [x, y]^\top$.

Question J.1: Let $\mathbf{z}_u = [0, 1]^\top$. Where could node v be embedded if the decoder evaluates to $\mathrm{DEC}(\mathbf{z}_u, \mathbf{z}_v) = 0.25$?

A ☐ On a circle of radius 0.5 centred at $\mathbf{z}_u$, i.e. $x^2 + (y-1)^2 = 0.25$
B ☐ On a circle of radius 1 centred at the origin, i.e. $x^2 + y^2 = 1$
C ☐ On the line $x = 0$
D ☐ On the line $y = 0$
E ☐ None of the above

Show the main steps of your derivation.

Exercise K — Message passing

Consider a star graph with centre a and leaves b, c, d (so $\mathcal{N}(a) = \{b, c, d\}$ and $\mathcal{N}(b) = \mathcal{N}(c) = \mathcal{N}(d) = \{a\}$). We use the message-passing GNN

$$h_u^{(k)} = w_0 + w_1 h_u^{(k-1)} + w_2 \sum_{v \in \mathcal{N}(u)} h_v^{(k-1)}, \tag{15}$$

with $w_0 = w_1 = w_2 = 1$ and initial features $h_u^{(0)} = 1$ for all u.

Question K.1: What are the node features after two message-passing rounds?

$$h_a^{(2)} = \qquad\qquad h_b^{(2)} = h_c^{(2)} = h_d^{(2)} =$$

Show the main steps of your derivation.

Question K.2: A single GNN layer can be written as the matrix–vector product $\mathbf{h}^{(k)} = \mathbf{w}_0 + M\,\mathbf{h}^{(k-1)}$, where $\mathbf{h}^{(k)} = [h_a^{(k)}, h_b^{(k)}, h_c^{(k)}, h_d^{(k)}]^\top$ and $\mathbf{w}_0 = [w_0, w_0, w_0, w_0]^\top$. Again with $w_1 = w_2 = 1$, fill in all non-zero entries of M.

$$M = \begin{bmatrix} \quad & \quad & \quad \\ & & \\ & & \end{bmatrix}$$